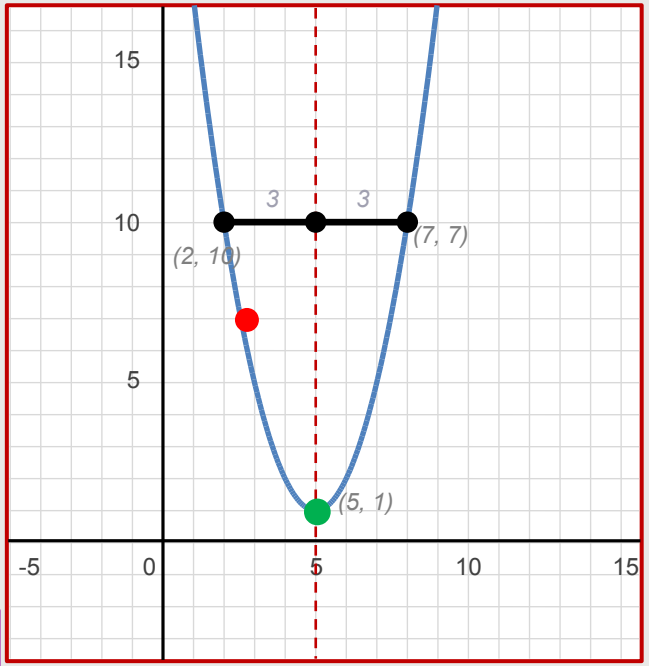
Convex Function, Learning Rate =0.2, Start at X=3



step	0.2
initial X	3
delta	0.1

$f(x) = (x-5)^2 + 1$

$X = 5$

f(x) Minimum value is **1.0**

Reach Minimum after 11 Steps

x	3	3.2	3.4	3.6	3.8	4	4.2	4.4	4.6	4.8	5
f(x)	5	4.24	3.56	2.96	2.44	2	1.64	1.36	1.16	1.04	1
x-delta	2.9	3.1	3.3	3.5	3.7	3.9	4.1	4.3	4.5	4.7	4.9
f(x-delta)	5.41	4.61	3.89	3.25	2.69	2.21	1.81	1.49	1.25	1.09	1.01
x+delta	3.1	3.3	3.5	3.7	3.9	4.1	4.3	4.5	4.7	4.9	5.1
f(x+delta)	4.61	3.89	3.25	2.69	2.21	1.81	1.49	1.25	1.09	1.01	1.01
slope	-8	-7.2	-6.4	-5.6	-4.8	-4	-3.2	-2.4	-1.6	-0.8	0
next x	3.2	3.4	3.6	3.8	4	4.2	4.4	4.6	4.8	5	stop